MARIELLE EDUARDO

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EDUCATION

Ph.D. Astronomy, University of Victoria — Sep 2022–present
Thesis Title: Exploring the Trans-Neptunian Space with James Webb Space Telescope
Supervisor* and Committee members: Dr. Wesley Fraser*, Dr. Ruobing Dong, Dr. JJ Kavelaars, Dr. Maria Womack

M.Sc. Astronomy, National Central University — Feb 2019–Jan 2021
Thesis Title: Hunting Trans-Neptunian Objects with the Hyper Suprime – Cam Subaru Strategic Program (HSC-SSP) Deep and UltraDeep Layers
Supervisor* and Committee members: Prof. Wen-Ping Chen*, Dr. Shiang-Yu Wang*, Dr. Wing-Huen Ip

B.Sc. Physics, University of the Philippines Baguio — Jun 2013–Jul 2018
Thesis Title: Correlation of the New Sunspot Number and Geomagnetic aa Index in the Years 1900-2013
Supervisor* and Committee members: Prof. Bhazel Pelicano*, Prof. Alladin Jasmin*, Dr. Quirino Sugon Jr., Dr. Reinabelle Reyes

CURRENT PROJECTS

1. Mapping the Colors of the Smallest KBOs with PANORAMIC: Utilizing data from the *Parallel Wide Area NIRCam Observations to Reveal and Measure the Invisible Cosmos (PANORAMIC)* survey to constrain the KBO luminosity function in the critically undersampled magnitude range of 26.5 < m < 28 and obtain the only known multi-band near-infrared color measurements of small KBOs.
2. Characterizing Flux Loss during the Ramp-Fitting step of JWST Pipeline: Investigating flux loss as a function of target motion during the JWST ramp-fitting process, with the goal of developing a customized, pipeline-integrated ramp-fitting algorithm to correct this systematic bias and improve photometric accuracy for fast-moving Solar System targets.

COLLABORATION MEMBERSHIPS

1. Vera C. Rubin Observatory Legacy Survey of Space and Time – Solar System Science Collaboration (LSST-SSSC)

PEER REVIEWED PUBLICATIONS (FIRST AUTHOR)

1. **Eduardo, M.**, Morgan, A., Fraser, W., Trilling, D., Bernstein, G., Holman, M., Stansberry, J., HIlbert, B., Grundy, W. and Napier, K., 2026. The Luminosity Function of Ultra-Faint Trans-Neptunian Objects Detected by James Webb Space Telescope. *The Astronomical Journal*, accepted.

PEER REVIEWED PUBLICATIONS (CO-AUTHOR)

1. Morgan, A., **Eduardo, M.**, Trilling, D., Fraser, W., Stansberry, J., Bernstein, G., Hilbert, B., Holman, M., Grundy, W., Tegler, S. and Fuentes, C., 2026. Combined JWST and HST Deep Imaging to Characterize the Smallest Known TNOs. *The Astronomical Journal*, accepted.
2. Côté, P., Woods, T.E., Hutchings, J.B., Rhodes, J.D., Sánchez-Janssen, R., Scott, A.D., Pazder, J., Amenouche, M., Balogh, M., Blouin, S., Cournoyer, A., et al. (including **Eduardo, M.R.**), 2025. The CASTOR mission. *Journal of Astronomical Telescopes, Instruments, and Systems*, 11(4), 042202.
3. Chen, Y.T., **Eduardo, M.R.**, Muñoz-Gutiérrez, M.A., Wang, S.Y., Lehner, M.J. and Chang, C.K., 2022. A Lowinclination Neutral Trans-Neptunian Object in an Extreme Orbit. *The Astrophysical Journal Letters*, *937*(2), p.L22.
4. Alexandersen, M., Benecchi, S.D., Chen, Y.T., **Eduardo, M.R.**, Thirouin, A., Schwamb, M.E., Lehner, M.J., Wang, S.Y., Bannister, M.T., Gladman, B.J. and Gwyn, S.D., 2019. OSSOS. XII. Variability Studies of 65 Trans-Neptunian Objects Using the Hyper Suprime-Cam∗. *The Astrophysical Journal Supplement Series*, *244*(1), p.19.

CONFERENCE PROCEEDINGS

1. **Eduardo, M.R.,** Sugon, Q.M. and Pelicano, B.A.R., 2018. Correlation of the New Sunspot Number and Geomagnetic aa Index in the Years 1900-2013. *Proceedings of the International Astronomical Union*, *13*(S340), pp.43-46.

RESEARCH EXPERIENCE

1. Research Assistant, Academia Sinica Institute of Astronomy and Astrophysics — Feb 2021–Jul 2022
2. Summer Research Student, Academia Sinica Institute of Astronomy and Astrophysics — Jul 2017–Aug 2017
3. Summer Intern, Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA) Observatory — Jun 2016–Jul 2016

TEACHING AND MENTORSHIP

Research Co-Supervisor

Thomas Storer, Co-op project, NRC-Herzberg Astronomy and Astrophysics Research Centre — Jan 2026-Apr 2026

Mary Christalyn De Vera, Undergraduate Thesis, University of the Philippines — Feb 2024-May 2024

Teaching Assistant, University of Victoria

ASTR 250: Introduction to Astrophysics — Spring 2026

ASTR 201: The Search for Life in the Universe — Fall 2023 & 2025

ASTR 101: Exploring the Night Sky — Summer 2023, 2024 & 2025; Fall 2024; Spring 2025

PHYS 329: Intermediate Physics Laboratory — Fall 2024; Spring 2025; Summer 2025

ASTR 102: Exploring the Cosmos — Spring 2023 & 2024

ASTR 250: Introduction to Astrophysics — Fall 2024

PHYS 102A: General Physics — Fall 2022

Lecturer in Physics, University of the Philippines Baguio

PHYS 101.1: Fundamental Physics I Laboratory — Fall 2021

PHYS 131.1: Computational Physics Laboratory — Fall 2021

PHYS 195: Solar System Astrophysics — Fall 2021

CONTRIBUTED AND INVITED TALKS

☆ indicates invited talk, otherwise contributed

1. **Asteroids, Comets, Meteors Conference 2026**, Poznan, Poland — Jul 2026
2. **Canada Astronomical Society Annual Meeting**, Montreal, Canada — Jun 2026
3. **Europlanet Science Congress (EPSC)–Division of Planetary Science (DPS) Joint Meeting,** Helsinki, Finland — Sep 2025
4. **Progress in Understanding the Pluto System: 10 Years After Flyby,** Baltimore, USA — Jul 2025
5. **Trans-Neptunian Solar System 2024**, Taipei, Taiwan — Jun 2024
6. ☆**New Horizons Science Team Meeting**, Boulder, USA — Sep 2024
7. **Astronomical Society of Republic of China Annual Meeting**, Taipei, Taiwan — Sep 2021
8. **IAU Symposium 340: Long-term Datasets for the Understanding of Solar and Stellar Magnetic Cycles**, Jaipur, India — Feb 2018

CONFERENCE POSTER PRESENTATIONS

1. **Asteroids, Comets, Meteors Conference 2023,** Flagstaff, USA — Jun 2023
2. **Canada Astronomical Society Annual Meeting**, Penticton, Canada — Jun 2023
3. **Taiwan Physical Society Annual Meeting 2020**, Pingtung, Taiwan — Feb 2020

WORKSHOPS AND SCHOOLS

Rubin Data Academy 2025, Vera C. Rubin Observatory (remote) — Jul 2025

Summer School in Statistics for Astronomers, Pennsylvania State University (remote) — Jun 2024

Intermediate Python for Astronomical Software Development, LSST-TVS Software Task Force (remote) — Jul 2024

6th Bryurakan International Summer School, Byurakan, Armenia — Sep 2018

SUCCESSFUL OBSERVING PROPOSALS

1. **Exploring the Colors and Lightcurve of a Potential Contact Binary TNO**, Principal Investigator — Gemini-N 2025 A
   - Granted a 3.47-hour observation through the Gemini-North Fast Turnaround (FT) program to study a potential contact binary TNO identified in our HST and JWST data.
2. **The Supremely Deep TNO survey: A critical test of planet formation models**, Co-Investigator — JWST Cycle 4 GO
   - Awarded 135.1 hours to carry out a supremely deep program focused on measuring the size distribution of the smallest cold classical TNOs (down to 1 km in diameter) to place strong constraints on the processes of planet formation

SCHOLARSHIPS AND AWARDS

Canadian Space Agency Conference Travel Grant, $3,628 — 2026

Dr. Margaret "Marmie" Perkins Hess Graduate Fellowship, 2x $15,000 — 2024 & 2025

University of Victoria Graduate Fellowship (Doctoral), $7,240 — 2025

University of Victoria Graduate Fellowship (Doctoral), $6,207 — 2024

University of Victoria Graduate Fellowship (Doctoral), 2x $5,000 — 2024 & 2025

Taiwan Astronomical Society of the Republic of China Annual Meeting Best Poster Presentation Award, $NT3,000 — 2020